

## OPCODES TABLE OF INTEL 8085

### Opcodes of Intel 8085 in Alphabetical Order

| Sr. No. | Mnemonics, Operand | Opcode | Bytes |
|---------|--------------------|--------|-------|
| 1.      | ACI Data           | CE     | 2     |
| 2.      | ADC A              | 8F     | 1     |
| 3.      | ADC B              | 88     | 1     |
| 4.      | ADC C              | 89     | 1     |
| 5.      | ADC D              | 8A     | 1     |
| 6.      | ADC E              | 8B     | 1     |
| 7.      | ADC H              | 8C     | 1     |
| 8.      | ADC L              | 8D     | 1     |
| 9.      | ADC M              | 8E     | 1     |
| 10.     | ADD A              | 87     | 1     |
| 11.     | ADD B              | 80     | 1     |
| 12.     | ADD C              | 81     | 1     |
| 13.     | ADD D              | 82     | 1     |
| 14.     | ADD E              | 83     | 1     |
| 15.     | ADD H              | 84     | 1     |
| 16.     | ADD L              | 85     | 1     |
| 17.     | ADD M              | 86     | 1     |
| 18.     | ADI Data           | C6     | 2     |
| 19.     | ANA A              | A7     | 1     |
| 20.     | ANA B              | A0     | 1     |
| 21.     | ANA C              | A1     | 1     |
| 22.     | ANA D              | A2     | 1     |
| 23.     | ANA E              | A3     | 1     |
| 24.     | ANA H              | A4     | 1     |
| 25.     | ANA L              | A5     | 1     |
| 26.     | ANA M              | A6     | 1     |
| 27.     | ANI Data           | E6     | 2     |
| 28.     | CALL Label         | CD     | 3     |
| 29.     | CC Label           | DC     | 3     |
| 30.     | CM Label           | FC     | 3     |
| 31.     | CMA                | 2F     | 1     |
| 32.     | CMC                | 3F     | 1     |
| 33.     | CMP A              | BF     | 1     |
| 34.     | CMP B              | B8     | 1     |
| 35.     | CMP C              | B9     | 1     |
| 36.     | CMP D              | BA     | 1     |
| 37.     | CMP E              | BB     | 1     |
| 38.     | CMP H              | BC     | 1     |
| 39.     | CMP L              | BD     | 1     |
| 40.     | CMP M              | BD     | 1     |
| 41.     | CNC Label          | D4     | 3     |
| 42.     | CNZ Label          | C4     | 3     |

| Sr. No. | Mnemonics, Operand | Opcode | Bytes |
|---------|--------------------|--------|-------|
| 43.     | CP Label           | F4     | 3     |
| 44.     | CPE Label          | EC     | 3     |
| 45.     | CPI Data           | FE     | 2     |
| 46.     | CPO Label          | E4     | 3     |
| 47.     | CZ Label           | CC     | 3     |
| 48.     | DAA                | 27     | 1     |
| 49.     | DAD B              | 09     | 1     |
| 50.     | DAD D              | 19     | 1     |
| 51.     | DAD H              | 29     | 1     |
| 52.     | DAD SP             | 39     | 1     |
| 53.     | DCR A              | 3D     | 1     |
| 54.     | DCR B              | 05     | 1     |
| 55.     | DCR C              | 0D     | 1     |
| 56.     | DCR D              | 15     | 1     |
| 57.     | DCR E              | 1D     | 1     |
| 58.     | DCR H              | 25     | 1     |
| 59.     | DCR L              | 2D     | 1     |
| 60.     | DCR M              | 35     | 1     |
| 61.     | DCX B              | 0B     | 1     |
| 62.     | DCX D              | 1B     | 1     |
| 63.     | DCX H              | 2B     | 1     |
| 64.     | DCX SP             | 3B     | 1     |
| 65.     | DI                 | F3     | 1     |
| 66.     | EI                 | FB     | 1     |
| 67.     | HLT                | 76     | 1     |
| 68.     | IN Port-address    | DB     | 2     |
| 69.     | INR A              | 3C     | 1     |
| 70.     | INR B              | 04     | 1     |
| 71.     | INR C              | 0C     | 1     |
| 72.     | INR D              | 14     | 1     |
| 73.     | INR E              | 1C     | 1     |
| 74.     | INR H              | 24     | 1     |
| 75.     | INR L              | 2C     | 1     |
| 76.     | INR M              | 34     | 1     |
| 77.     | INX B              | 03     | 1     |
| 78.     | INX D              | 13     | 1     |
| 79.     | INX H              | 23     | 1     |
| 80.     | INX SP             | 33     | 1     |
| 81.     | JC Label           | DA     | 3     |
| 82.     | JM Label           | FA     | 3     |
| 83.     | JMP Label          | C3     | 3     |
| 84.     | JNC Label          | D2     | 3     |
| 85.     | JNZ Label          | C2     | 3     |
| 86.     | JP Label           | F2     | 3     |
| 87.     | JPE Label          | EA     | 3     |
| 88.     | JPO Label          | E2     | 3     |
| 89.     | JZ Label           | CA     | 3     |

| Sr. No. | Mnemonics, Operand | Opcode | Bytes |
|---------|--------------------|--------|-------|
| 90.     | LDA Address        | 3A     | 3     |
| 91.     | LDAX B             | 0A     | 1     |
| 92.     | LDAX D             | 1A     | 1     |
| 93.     | LHLD Address       | 2A     | 3     |
| 94.     | LXI B              | 01     | 3     |
| 95.     | LXI D              | 11     | 3     |
| 96.     | LXI H              | 21     | 3     |
| 97.     | LXI SP             | 31     | 3     |
| 98.     | MOV A, A           | 7F     | 1     |
| 99.     | MOV A, B           | 78     | 1     |
| 100.    | MOV A, C           | 79     | 1     |
| 101.    | MOV A, D           | 7A     | 1     |
| 102.    | MOV A, E           | 7B     | 1     |
| 103.    | MOV A, H           | 7C     | 1     |
| 104.    | MOV A, L           | 7D     | 1     |
| 105.    | MOV A, M           | 7E     | 1     |
| 106.    | MOV B, A           | 47     | 1     |
| 107.    | MOV B, B           | 40     | 1     |
| 108.    | MOV B, C           | 41     | 1     |
| 109.    | MOV B, D           | 42     | 1     |
| 110.    | MOV B, E           | 43     | 1     |
| 111.    | MOV B, H           | 44     | 1     |
| 112.    | MOV B, L           | 45     | 1     |
| 113.    | MOV B, M           | 46     | 1     |
| 114.    | MOV C, A           | 4F     | 1     |
| 115.    | MOV C, B           | 48     | 1     |
| 116.    | MOV C, C           | 49     | 1     |
| 117.    | MOV C, D           | 4A     | 1     |
| 118.    | MOV C, E           | 4B     | 1     |
| 119.    | MOV C, H           | 4C     | 1     |
| 120.    | MOV C, L           | 4D     | 1     |
| 121.    | MOV C, M           | 4E     | 1     |
| 122.    | MOV D, A           | 57     | 1     |
| 123.    | MOV D, B           | 50     | 1     |
| 124.    | MOV D, C           | 51     | 1     |
| 125.    | MOV D, D           | 52     | 1     |
| 126.    | MOV D, E           | 53     | 1     |
| 127.    | MOV D, H           | 54     | 1     |
| 128.    | MOV D, L           | 55     | 1     |
| 129.    | MOV D, M           | 56     | 1     |
| 130.    | MOV E, A           | 5F     | 1     |
| 131.    | MOV E, B           | 58     | 1     |
| 132.    | MOV E, C           | 59     | 1     |
| 133.    | MOV E, D           | 5A     | 1     |
| 134.    | MOV E, E           | 5B     | 1     |
| 135.    | MOV E, H           | 5C     | 1     |
| 136.    | MOV E, L           | 5D     | 1     |

| Sr. No. | Mnemonics, Operand | Opcode | Bytes |
|---------|--------------------|--------|-------|
| 137.    | MOV E, M           | 5E     | 1     |
| 138.    | MOV H, A           | 67     | 1     |
| 139.    | MOV H, B           | 60     | 1     |
| 140.    | MOV H, C           | 61     | 1     |
| 141.    | MOV H, D           | 62     | 1     |
| 142.    | MOV H, E           | 63     | 1     |
| 143.    | MOV H, H           | 64     | 1     |
| 144.    | MOV H, L           | 65     | 1     |
| 145.    | MOV H, M           | 66     | 1     |
| 146.    | MOV L, A           | 6F     | 1     |
| 147.    | MOV L, B           | 68     | 1     |
| 148.    | MOV L, C           | 69     | 1     |
| 149.    | MOV L, D           | 6A     | 1     |
| 150.    | MOV L, E           | 6B     | 1     |
| 151.    | MOV L, H           | 6C     | 1     |
| 152.    | MOV L, L           | 6D     | 1     |
| 153.    | MOV L, M           | 6E     | 1     |
| 154.    | MOV M, A           | 77     | 1     |
| 155.    | MOV M, B           | 70     | 1     |
| 156.    | MOV M, C           | 71     | 1     |
| 157.    | MOV M, D           | 72     | 1     |
| 158.    | MOV M, E           | 73     | 1     |
| 159.    | MOV M, H           | 74     | 1     |
| 160.    | MOV M, L           | 75     | 1     |
| 161.    | MVI A, Data        | 3E     | 2     |
| 162.    | MVI B, Data        | 06     | 2     |
| 163.    | MVI C, Data        | 0E     | 2     |
| 164.    | MVI D, Data        | 16     | 2     |
| 165.    | MVI E, Data        | 1E     | 2     |
| 166.    | MVI H, Data        | 26     | 2     |
| 167.    | MVI L, Data        | 2E     | 2     |
| 168.    | MVI M, Data        | 36     | 2     |
| 169.    | NOP                | 00     | 1     |
| 170.    | ORA A              | B7     | 1     |
| 171.    | ORA B              | B0     | 1     |
| 172.    | ORA C              | B1     | 1     |
| 173.    | ORA D              | B2     | 1     |
| 174.    | ORA E              | B3     | 1     |
| 175.    | ORA H              | B4     | 1     |
| 176.    | ORA L              | B5     | 1     |
| 177.    | ORA M              | B6     | 1     |
| 178.    | ORI Data           | F6     | 2     |
| 179.    | OUT Port-Address   | D3     | 2     |
| 180.    | PCHL               | E9     | 1     |
| 181.    | POP B              | C1     | 1     |
| 182.    | POP D              | D1     | 1     |
| 183.    | POP H              | E1     | 1     |

| Sr. No. | Mnemonics, Operand | Opcode | Bytes |
|---------|--------------------|--------|-------|
| 184.    | POP PSW            | F1     | 1     |
| 185.    | PUSH B             | C5     | 1     |
| 186.    | PUSH D             | D5     | 1     |
| 187.    | PUSH H             | E5     | 1     |
| 188.    | PUSH PSW           | F5     | 1     |
| 189.    | RAL                | 17     | 1     |
| 190.    | RAR                | 1F     | 1     |
| 191.    | RC                 | D8     | 1     |
| 192.    | RET                | C9     | 1     |
| 193.    | RIM                | 20     | 1     |
| 194.    | RLC                | 07     | 1     |
| 195.    | RM                 | F8     | 1     |
| 196.    | RNC                | D0     | 1     |
| 197.    | RNZ                | C0     | 1     |
| 198.    | RP                 | F0     | 1     |
| 199.    | RPE                | E8     | 1     |
| 200.    | RPO                | E0     | 1     |
| 201.    | RRC                | 0F     | 1     |
| 202.    | RST 0              | C7     | 1     |
| 203.    | RST 1              | CF     | 1     |
| 204.    | RST 2              | D7     | 1     |
| 205.    | RST 3              | DF     | 1     |
| 206.    | RST 4              | E7     | 1     |
| 207.    | RST 5              | EF     | 1     |
| 208.    | RST 6              | F7     | 1     |
| 209.    | RST 7              | FF     | 1     |
| 210.    | RZ                 | C8     | 1     |
| 211.    | SBB A              | 9F     | 1     |
| 212.    | SBB B              | 98     | 1     |
| 213.    | SBB C              | 99     | 1     |
| 214.    | SBB D              | 9A     | 1     |
| 215.    | SBB E              | 9B     | 1     |
| 216.    | SBB H              | 9C     | 1     |
| 217.    | SBB L              | 9D     | 1     |
| 218.    | SBB M              | 9E     | 1     |
| 219.    | SBI Data           | DE     | 2     |
| 220.    | SHLD Address       | 22     | 3     |
| 221.    | SIM                | 30     | 1     |
| 222.    | SPHL               | F9     | 1     |
| 223.    | STA Address        | 32     | 3     |
| 224.    | STAX B             | 02     | 1     |
| 225.    | STAX D             | 12     | 1     |
| 226.    | STC                | 37     | 1     |
| 227.    | SUB A              | 97     | 1     |
| 228.    | SUB B              | 90     | 1     |
| 229.    | SUB C              | 91     | 1     |
| 230.    | SUB D              | 92     | 1     |

| Sr. No. | Mnemonics, Operand | Opcode | Bytes |
|---------|--------------------|--------|-------|
| 231.    | SUB E              | 93     | 1     |
| 232.    | SUB H              | 94     | 1     |
| 233.    | SUB L              | 95     | 1     |
| 234.    | SUB M              | 96     | 1     |
| 235.    | SUI Data           | D6     | 2     |
| 236.    | XCHG               | EB     | 1     |
| 237.    | XRA A              | AF     | 1     |
| 238.    | XRA B              | A8     | 1     |
| 239.    | XRA C              | A9     | 1     |
| 240.    | XRA D              | AA     | 1     |
| 241.    | XRA E              | AB     | 1     |
| 242.    | XRA H              | AC     | 1     |
| 243.    | XRA L              | AD     | 1     |
| 244.    | XRA M              | AE     | 1     |
| 245.    | XRI Data           | EE     | 2     |
| 246.    | XTHL               | E3     | 1     |